

ACR42606

## Tebbe reagent, 0.5M solution in toluene

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

|                                                                 |                                                                                                                                                                                                                                                                    |
|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>产品说明:</b><br><b>Product Description:</b>                     | <b>氯化双(5-茂二甲基铝-甲基钛), 0.5M甲苯溶液</b><br><b>Tebbe reagent, 0.5M solution in toluene</b>                                                                                                                                                                                |
| <b>Cat No. :</b><br><b>Synonyms</b><br><b>Molecular Formula</b> | <b>426060000; 426061000</b><br>Bis(cyclopentadienyl-delta-chloro(dimethylaluminium-delta-methylenetitanium<br>C13 H18 Al Cl Ti                                                                                                                                     |
| <b>Supplier</b>                                                 | <b>UK entity/business name</b><br>Fisher Scientific UK<br>Bishop Meadow Road,<br>Loughborough, Leicestershire LE11 5RG, United Kingdom<br><br><b>EU entity/business name</b><br>Thermo Fisher Scientific<br>Janssen Pharmaceuticaaan 3a, 2440 Geel, Belgium        |
| <b>Emergency Telephone Number</b>                               | For information <b>US</b> call: 001-800-227-6701 / <b>Europe</b> call: +32 14 57 52 11<br>Emergency Number <b>US</b> :001-201-796-7100 / <b>Europe</b> : +32 14 57 52 99<br><b>CHEMTREC</b> Tel. No. <b>US</b> :001-800-424-9300 / <b>Europe</b> :001-703-527-3887 |
| <b>E-mail address</b>                                           | begel.sdsdesk@thermofisher.com                                                                                                                                                                                                                                     |
| <b>Recommended Use</b><br><b>Uses advised against</b>           | Laboratory chemicals.<br>No Information available                                                                                                                                                                                                                  |

### SECTION 2. HAZARD IDENTIFICATION

|                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                      |                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|-----------------------------------------|
| <b>Physical State</b><br>Liquid                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Appearance</b><br>Dark red Purple | <b>Odor</b><br>No information available |
| <b>Emergency Overview</b><br>Highly flammable liquid and vapor. May be fatal if swallowed and enters airways. Suspected of damaging fertility or the unborn child. May cause drowsiness and dizziness. Toxic to aquatic life. Harmful to aquatic life with long lasting effects. Causes severe skin burns and eye damage. May cause damage to organs through prolonged or repeated exposure. Moisture sensitive. Air sensitive. |                                      |                                         |

#### Classification of the substance or mixture

|                                                      |              |
|------------------------------------------------------|--------------|
| Flammable liquids.                                   | Category 2   |
| Aspiration Toxicity                                  | Category 1   |
| Skin Corrosion/Irritation                            | Category 1 B |
| Serious Eye Damage/Eye Irritation                    | Category 1   |
| Reproductive Toxicity                                | Category 2   |
| Specific target organ toxicity - (single exposure)   | Category 3   |
| Specific target organ toxicity - (repeated exposure) | Category 2   |
| Acute aquatic toxicity                               | Category 2   |
| Chronic aquatic toxicity                             | Category 3   |

#### Label Elements

# SAFETY DATA SHEET

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**Signal Word****Danger****Hazard Statements**

H225 - Highly flammable liquid and vapor  
H304 - May be fatal if swallowed and enters airways  
H336 - May cause drowsiness or dizziness  
H401 - Toxic to aquatic life  
H412 - Harmful to aquatic life with long lasting effects  
H314 - Causes severe skin burns and eye damage  
H373 - May cause damage to organs through prolonged or repeated exposure  
H361 - Suspected of damaging fertility or the unborn child

**Precautionary Statements****Prevention**

P201 - Obtain special instructions before use  
P202 - Do not handle until all safety precautions have been read and understood  
P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P240 - Ground and bond container and receiving equipment  
P242 - Use non-sparking tools  
P243 - Take action to prevent static discharges  
P260 - Do not breathe dust/fume/gas/mist/vapors/spray  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

**Response**

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P310 - Immediately call a POISON CENTER or doctor  
P330 - Rinse mouth  
P331 - Do NOT induce vomiting  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
P362 + P364 - Take off contaminated clothing and wash it before reuse

**Storage**

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

Vapors may cause flash fire or explosion. Highly flammable. Water reactive.

**Health Hazards**

Aspiration hazard if swallowed - can enter lungs and cause damage. Suspected of damaging fertility or the unborn child. May cause drowsiness or dizziness. Corrosive. Causes skin and eye burns. May cause damage to organs through prolonged or repeated exposure.

**Environmental hazards**

Toxic to aquatic life. Harmful to aquatic life with long lasting effects. . Is not likely mobile in the environment. Reacts with water.

**Other Hazards**

Toxic to terrestrial vertebrates. This product does not contain any known or suspected endocrine disruptors.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

# SAFETY DATA SHEET

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| Component                                                                                    | CAS No     | Weight % |
|----------------------------------------------------------------------------------------------|------------|----------|
| Toluene                                                                                      | 108-88-3   | 85       |
| Titanium,<br>.mu.-chlorobis(eta.5-2,4-cyclopentadien-1-yl)(dimethylaluminum)-.mu.-methylene- | 67719-69-1 | 15       |

## SECTION 4. FIRST AID MEASURES

### General Advice

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.

### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Call a physician immediately.

### Inhalation

If not breathing, give artificial respiration. Remove from exposure, lie down. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Call a physician immediately. Risk of serious damage to the lungs (by aspiration).

### Ingestion

Do NOT induce vomiting. Clean mouth with water. Never give anything by mouth to an unconscious person. Call a physician immediately. Call a physician or poison control center immediately. If vomiting occurs naturally, have victim lean forward.

### Most important symptoms and effects

Causes burns by all exposure routes. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

### Self-Protection of the First Aider

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

### Notes to Physician

Treat symptomatically. Symptoms may be delayed.

## SECTION 5. FIRE-FIGHTING MEASURES

### Suitable Extinguishing Media

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam. Water mist may be used to cool closed containers.

### Extinguishing media which must not be used for safety reasons

No information available.

### Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 6. ACCIDENTAL RELEASE MEASURES

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### Personal Precautions

Use personal protective equipment as required. Ensure adequate ventilation. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Remove all sources of ignition. Take precautionary measures against static discharges.

### Environmental Precautions

Do not flush into surface water or sanitary sewer system.

### Methods for Containment and Clean Up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7. HANDLING AND STORAGE

### Handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Handle under an inert atmosphere. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges.

### Storage

Corrosives area. Flammables area. Keep away from heat, sparks and flame. Keep under nitrogen. Store in freezer. Store under an inert atmosphere. Protect from moisture. Keep containers tightly closed in a dry, cool and well-ventilated place.

### Specific Use(s)

Use in laboratories

## SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component | China                                                            | Taiwan                                     | Thailand                                          | Hong Kong                                 |
|-----------|------------------------------------------------------------------|--------------------------------------------|---------------------------------------------------|-------------------------------------------|
| Toluene   | TWA: 50 mg/m <sup>3</sup><br>STEL: 100 mg/m <sup>3</sup><br>Skin | TWA: 100 ppm<br>TWA: 376 mg/m <sup>3</sup> | Ceiling: 300 ppm<br>STEL: 500 ppm<br>TWA: 200 ppm | TWA: 50 ppm<br>TWA: 188 mg/m <sup>3</sup> |

| Component | ACGIH TLV   | OSHA PEL                                                                                                                                                               | NIOSH                                                                                                       | The United Kingdom                                                                                                        | European Union                                                                                                                |
|-----------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Toluene   | TWA: 20 ppm | (Vacated) TWA: 100 ppm<br>(Vacated) TWA: 375 mg/m <sup>3</sup><br>Ceiling: 300 ppm<br>(Vacated) STEL: 150 ppm<br>(Vacated) STEL: 560 mg/m <sup>3</sup><br>TWA: 200 ppm | IDLH: 500 ppm<br>TWA: 100 ppm<br>TWA: 375 mg/m <sup>3</sup><br>STEL: 150 ppm<br>STEL: 560 mg/m <sup>3</sup> | STEL: 100 ppm 15 min<br>STEL: 384 mg/m <sup>3</sup> 15 min<br>TWA: 50 ppm 8 hr<br>TWA: 191 mg/m <sup>3</sup> 8 hr<br>Skin | TWA: 50 ppm (8hr)<br>TWA: 192 mg/m <sup>3</sup> (8hr)<br>STEL: 100 ppm (15min)<br>STEL: 384 mg/m <sup>3</sup> (15min)<br>Skin |

### Legend

ACGIH - American Conference of Governmental Industrial Hygienists  
 OSHA - Occupational Safety and Health Administration  
 NIOSH: NIOSH - National Institute for Occupational Safety and Health

### Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS70 General methods for sampling airborne gases and vapours MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas

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chromatography

### Exposure Controls

#### Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting equipment. Ensure adequate ventilation, especially in confined areas. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

#### Personal protective equipment

**Eye Protection** Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material              | Breakthrough time                    | Glove thickness | EU standard | Glove comments        |
|-----------------------------|--------------------------------------|-----------------|-------------|-----------------------|
| Nitrile rubber<br>Viton (R) | See manufacturers<br>recommendations | -               | EN 374      | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Long sleeved clothing

**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** low boiling organic solvent Type AX Brown conforming to EN371 or Organic gases and vapours filter Type A Brown conforming to EN14387

**Small scale/Laboratory use** Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141  
When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** Prevent product from entering drains. Do not allow material to contaminate ground water system.

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

**Appearance** Dark red Purple  
**Physical State** Liquid

**Odor** No information available  
**Odor Threshold** No data available  
**pH** Not applicable  
**Melting Point/Range** No data available  
**Softening Point** No data available  
**Boiling Point/Range** No information available

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|                                                |                          |                                             |
|------------------------------------------------|--------------------------|---------------------------------------------|
| <b>Flash Point</b>                             | 4 °C / 39.2 °F           | <b>Method</b> - No information available    |
| <b>Evaporation Rate</b>                        | No data available        |                                             |
| <b>Flammability (solid,gas)</b>                | Not applicable           | Liquid                                      |
| <b>Explosion Limits</b>                        | No data available        |                                             |
| <b>Vapor Pressure</b>                          | No data available        |                                             |
| <b>Vapor Density</b>                           | No data available        | (Air = 1.0)                                 |
| <b>Specific Gravity / Density</b>              | 0.927                    |                                             |
| <b>Bulk Density</b>                            | Not applicable           | Liquid                                      |
| <b>Water Solubility</b>                        | Reacts with water        |                                             |
| <b>Solubility in other solvents</b>            | No information available |                                             |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |                                             |
| <b>Component</b>                               | <b>log Pow</b>           |                                             |
| Toluene                                        | 2.73                     |                                             |
| <b>Autoignition Temperature</b>                | No data available        |                                             |
| <b>Decomposition Temperature</b>               | No data available        |                                             |
| <b>Viscosity</b>                               | No data available        |                                             |
| <b>Explosive Properties</b>                    |                          | Vapors may form explosive mixtures with air |
| <b>Oxidizing Properties</b>                    | No information available |                                             |
| <b>Molecular Formula</b>                       | C13 H18 Al Cl Ti         |                                             |
| <b>Molecular Weight</b>                        | 284.62                   |                                             |

## SECTION 10. STABILITY AND REACTIVITY

|                                 |                                                                                                                                                                              |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Stability</b>                | Moisture sensitive. Air sensitive.                                                                                                                                           |
| <b>Hazardous Reactions</b>      | None under normal processing.                                                                                                                                                |
| <b>Hazardous Polymerization</b> | Hazardous polymerization does not occur.                                                                                                                                     |
| <b>Conditions to Avoid</b>      | Incompatible products. Excess heat. Keep away from open flames, hot surfaces and sources of ignition. Exposure to moisture. Exposure to air. Exposure to moist air or water. |
| <b>Materials to avoid</b>       | Strong oxidizing agents.                                                                                                                                                     |

**Hazardous Decomposition Products** Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Hydrogen chloride gas.

## SECTION 11. TOXICOLOGICAL INFORMATION

### Product Information

(a) acute toxicity;  
 Toxicology data for the components

| Component | LD50 Oral            | LD50 Dermal            | LC50 Inhalation       |
|-----------|----------------------|------------------------|-----------------------|
| Toluene   | > 5000 mg/kg ( Rat ) | 12000 mg/kg ( Rabbit ) | 26700 ppm ( Rat ) 1 h |

(b) skin corrosion/irritation; Category 1 B

(c) serious eye damage/irritation; Category 1

(d) respiratory or skin sensitization;  
 Respiratory No data available  
 Skin No data available

(e) germ cell mutagenicity; No data available

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(f) carcinogenicity; No data available  
There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; Category 2  
Teratogenicity Possible risk of harm to the unborn child.

(h) STOT-single exposure; Category 3  
Results / Target organs Central nervous system (CNS)

(i) STOT-repeated exposure; Category 2  
Target Organs Neuropsychological effects, Eyes, Ears.

(j) aspiration hazard; Category 1

Symptoms / effects, both acute and delayed Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

## SECTION 12. ECOLOGICAL INFORMATION

Ecotoxicity effects The product contains following substances which are hazardous for the environment. Contains a substance which is: Toxic to aquatic organisms. Reacts with water so no ecotoxicity data for the substance is available.

| Component | Freshwater Fish                                                                                              | Water Flea                                                                                         | Freshwater Algae                                                                                                                 | Microtox                |
|-----------|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| Toluene   | 50-70 mg/L LC50 96 h<br>5-7 mg/L LC50 96 h<br>15-19 mg/L LC50 96 h<br>28 mg/L LC50 96 h<br>12 mg/L LC50 96 h | EC50: = 11.5 mg/L, 48h<br>(Daphnia magna)<br>EC50: 5.46 - 9.83 mg/L,<br>48h Static (Daphnia magna) | EC50: = 12.5 mg/L, 72h static<br>(Pseudokirchneriella subcapitata)<br>EC50: > 433 mg/L, 96h<br>(Pseudokirchneriella subcapitata) | EC50 = 19.7 mg/L 30 min |

### Persistence and Degradability

Persistence Persistence is unlikely.  
Degradability Reacts with water.

| Component                  | Degradability |
|----------------------------|---------------|
| Toluene<br>108-88-3 ( 85 ) | 86% (20d)     |

Degradation in sewage treatment plant Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants. Water reactive.

Bioaccumulative Potential Bioaccumulation is unlikely

| Component | log Pow | Bioconcentration factor (BCF) |
|-----------|---------|-------------------------------|
| Toluene   | 2.73    | 90                            |

Mobility in soil Reacts with water Is not likely mobile in the environment

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|                                        |                                                                           |
|----------------------------------------|---------------------------------------------------------------------------|
| <b>Endocrine Disruptor Information</b> | This product does not contain any known or suspected endocrine disruptors |
| <b>Persistent Organic Pollutant</b>    | This product does not contain any known or suspected substance            |
| <b>Ozone Depletion Potential</b>       | This product does not contain any known or suspected substance            |

## SECTION 13. DISPOSAL CONSIDERATIONS

|                                            |                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from Residues/Unused Products</b> | Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.                                                                                                                                                                     |
| <b>Contaminated Packaging</b>              | Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.                                                                                                   |
| <b>Other Information</b>                   | Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms. Do not let this chemical enter the environment. |

## SECTION 14. TRANSPORT INFORMATION

### Road and Rail Transport

|                                |                                                                                      |
|--------------------------------|--------------------------------------------------------------------------------------|
| <b>UN-No</b>                   | UN2924                                                                               |
| <b>Proper Shipping Name</b>    | Flammable liquid, corrosive, n.o.s.                                                  |
| <b>Technical Shipping Name</b> | Toluene, Bis(cyclopentadienyl-delta-chloro(dimethylaluminium-delta-methylenetitanium |
| <b>Hazard Class</b>            | 3                                                                                    |
| <b>Subsidiary Hazard Class</b> | 8                                                                                    |
| <b>Packing Group</b>           | II                                                                                   |

### IMDG/IMO

|                                |                                                                                      |
|--------------------------------|--------------------------------------------------------------------------------------|
| <b>UN-No</b>                   | UN2924                                                                               |
| <b>Proper Shipping Name</b>    | Flammable liquid, corrosive, n.o.s.                                                  |
| <b>Technical Shipping Name</b> | Toluene, Bis(cyclopentadienyl-delta-chloro(dimethylaluminium-delta-methylenetitanium |
| <b>Hazard Class</b>            | 3                                                                                    |
| <b>Subsidiary Hazard Class</b> | 8                                                                                    |
| <b>Packing Group</b>           | II                                                                                   |

### IATA

|                                |                                                                                      |
|--------------------------------|--------------------------------------------------------------------------------------|
| <b>UN-No</b>                   | UN2924                                                                               |
| <b>Proper Shipping Name</b>    | Flammable liquid, corrosive, n.o.s.                                                  |
| <b>Technical Shipping Name</b> | Toluene, Bis(cyclopentadienyl-delta-chloro(dimethylaluminium-delta-methylenetitanium |
| <b>Hazard Class</b>            | 3                                                                                    |
| <b>Subsidiary Hazard Class</b> | 8                                                                                    |
| <b>Packing Group</b>           | II                                                                                   |

|                                     |                                 |
|-------------------------------------|---------------------------------|
| <b>Special Precautions for User</b> | No special precautions required |
|-------------------------------------|---------------------------------|

## SECTION 15. REGULATORY INFORMATION

### **International Inventories**

X = listed, U.S.A. (TSCA), Canada (DSL/NDL), Europe (EINECS/ELINCS/NLP), Australia (AICS), Korea (KECL), China (IECSC), Japan (ENCS), Philippines (PICCS).

| Component | The Inventory of Hazardous Chemicals | List of dangerous goods GB 12268 - | TCSI | IECSC | EINECS | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL |
|-----------|--------------------------------------|------------------------------------|------|-------|--------|------|-----|-------|------|------|------|------|
|           |                                      |                                    |      |       |        |      |     |       |      |      |      |      |

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## Tebbe reagent, 0.5M solution in toluene

|                                                                                                           | (2015 Edition) | 2012 |   |   |           |   |   |   |   |   |   |          |
|-----------------------------------------------------------------------------------------------------------|----------------|------|---|---|-----------|---|---|---|---|---|---|----------|
| Toluene                                                                                                   | X              | X    | X | X | 203-625-9 | X | X | X | X | X | X | KE-33936 |
| Titanium,<br>.mu.-chlorobis(.eta.5-2<br>,4-cyclopentadien-1-yl)<br>(dimethylaluminum)-.m<br>u.-methylene- | -              | -    | X | - | -         | - | - | - | - |   | - | -        |

### National Regulations

## SECTION 16. OTHER INFORMATION

**Creation Date** 19-May-2009  
**Revision Date** 11-Apr-2024  
**Revision Summary** SDS sections updated.

### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

Chemical incident response training.

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadviser - LOLI, Merck index, RTECS

**Physical hazards**

On basis of test data

**Health Hazards**

Calculation method

**Environmental hazards**

Calculation method

# SAFETY DATA SHEET

Tebbe reagent, 0.5M solution in toluene

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## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**